

Shambhavi Goyal

📍 Buffalo, NY ✉ shambhavigoyal05@gmail.com ☎ +1 (716)580-0656 📁 Portfolio in LinkedIn 🐙 Github

Education

University at Buffalo, The State University of New York

Aug 2023 – Dec 2026

BS in Computer Science, Honors College

- **GPA:** 3.87/4.0 | **Awards:** Dean's List, Women Empowerment Scholarship, University at Buffalo Flagship Scholarship
- **Coursework:** Distributed Systems, Operating Systems, System Programming, Computer Organization, Computer Vision & Image Processing, Data Structures & Algorithms, Probability, Linear Algebra
- **Leadership:** Member of the Leadership House, Women in Science and Engineering

Skills

Programming Languages: C, C++, Java, Python, JavaScript, TypeScript, Go, HTML/CSS, Kotlin

Frameworks & Libraries: React, Node.js, Express, Bootstrap, Flask, Django, TensorFlow, PyTorch, Socket.IO

Databases: PostgreSQL, MySQL, MongoDB

Tools & Technologies: Git, Docker, Kubernetes, AWS, REST APIs, WebSockets, CI/CD (GitHub Actions)

Operating Systems: Windows, Linux, MacOS

Work Experience

Software Developer Intern

Berkeley, CA (Remote)

Meta Layer Initiative

Aug 2025 – Present

- Built a decentralized social platform with an AI-powered Chrome extension using Node.js/Express, PostgreSQL, React, and DeepSeek AI for contextual web analysis.
- Implemented real-time messaging with WebSockets, secured authentication with multiple OAuth providers, and delivered a responsive glassmorphism UI, optimizing reliability by 30% and supporting 100+ active sessions.

Undergraduate Teaching Assistant – Algorithms & Complexity, Discrete Structures

Buffalo, NY

CSE Department

Aug 2024 – Dec 2025

- Led weekly recitations and 20+ office hours on algorithms (greedy, divide-and-conquer, dynamic programming, NP-completeness), improving average class grades by 12% through clear explanations and problem-solving support.
- Assessed proofs, assignments, and exams while designing practice exercises that bridged theory with coding applications, strengthening students' mastery of data structures, algorithm design, and computational thinking in real-world contexts.

Technical Projects

CrowdSense – Real-Time Campus Occupancy Tracker

github [🔗](#)

- Built and deployed a computer vision-driven system using Ultralytics YOLO and FastAPI to analyze live CCTV feeds and deliver <150 ms real-time occupancy analytics for campus spaces via a React Native + Expo mobile app.
- Engineered a scalable, privacy-first pipeline integrating Python, PyTorch, SQLAlchemy, PostgreSQL, and WebSockets, enabling concurrent multi-camera tracking, instant map updates, and extensibility to city-wide crowd and parking analytics.

Smart Expense Splitter – AI-Powered Bill Splitting App

github [🔗](#)

- Built a full-stack bill splitting app with React Native, Node.js/Express, TypeScript, and PostgreSQL, implementing JWT authentication and real-time balance calculations across multiple groups supporting 50+ concurrent users.
- Integrated OpenAI Vision API for receipt OCR achieving 95%+ accuracy and <2s processing time, reducing manual entry by 85%; implemented recurring expenses via cron jobs and real-time notifications with WebSockets for itemized splitting.

AI Syllabus Parser – Automated Calendar Integration Tool

github [🔗](#)

- Built a full-stack web application using Next.js, TypeScript, and OpenAI GPT-4o-mini, implementing intelligent PDF parsing and automated assignment extraction with 95%+ accuracy, reducing manual calendar entry time by 90%.
- Integrated Google Calendar API with OAuth 2.0 authentication and ICS export, enabling cross-platform synchronization with sub-200ms response times for event updates.

Pintos Operating System – Kernel-Level Threading & Scheduler Implementation

github [🔗](#)

- Implemented kernel-level threading, synchronization, and CPU scheduling for the Pintos OS, extending support for priority scheduling, nested priority donation, and a multilevel feedback queue (MLFQ) scheduler.
- Optimized the thread management and timer subsystem by replacing busy-waiting with event-driven sleep queues, ensuring efficient context switching and reliable synchronization across concurrent kernel threads.